

Topology First-Year Exam, January 2017, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that the set of functions $f : \mathbb{N} \rightarrow \mathbb{Z}$ that are eventually zero is countable. A function $f : \mathbb{N} \rightarrow \mathbb{Z}$ is called eventually zero if there is N such that $f(n) = 0$ for all $n \geq N$.
2. Let X be a connected metric space having more than one point. Can X be countable?
3. This problem has three parts.
 - (a) Show that every continuous map $f : [0, 1] \rightarrow [0, 1]$ has a fixed point. Is this true for continuous maps
 - (b) $f : (0, 1) \rightarrow (0, 1)$?
 - (c) $f : [0, 1] \rightarrow [0, 1)$?
4. This problem has two parts.
 - (a) Does there exist a continuous surjective map from the 2-sphere S^2 to the interval $(0, 1)$?
 - (b) Does there exist a continuous injective map from the 2-sphere S^2 to the interval $(0, 1)$?
5. Let $f : X \rightarrow X$ be a map of a compact metric space to itself that satisfies the following condition:
 $d(f(x), f(y)) < d(x, y)$ whenever $x \neq y$.
 - (a) Prove that f is continuous.
 - (b) Show that f has a fixed point and the fixed point is unique.

Answer the following with complete definitions or statements or short proofs.

6. State the Intermediate Value Theorem.
7. State the Cantor–Schröder–Bernstein Theorem.
8. Is every connected space path connected?
9. What is a basis of a topology? Does the set of all half-open intervals $\{(a, b], [c, d) \mid a < b, c < d\}$ form a basis of a topology on $\mathbb{R}$?
10. Is $\mathbb{R}^\omega$ connected in the uniform topology?